

# Residual-Aware Bayesian Clustering:

## Complete Derivations from First Principles

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# 1 Probability Densities and Likelihood

## 1.1 The Gaussian density

A  $d$ -dimensional Gaussian (normal) distribution with mean  $\mu \in \mathbb{R}^d$  and covariance  $\Sigma \in \mathbb{R}^{d \times d}$  (symmetric positive definite) has density

$$\mathcal{N}(x; \mu, \Sigma) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right). \quad (1)$$

- The normalizer  $(2\pi)^{d/2} |\Sigma|^{1/2}$  ensures  $\int_{\mathbb{R}^d} \mathcal{N}(x; \mu, \Sigma) dx = 1$ .
- The exponent  $-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)$  is a quadratic form measuring “distance” from  $\mu$  in the metric defined by  $\Sigma^{-1}$ . Points far from  $\mu$  (in the  $\Sigma$ -metric) get low density.
- The **precision matrix**  $\Sigma^{-1}$  controls the shape: large eigenvalues of  $\Sigma^{-1}$  (small eigenvalues of  $\Sigma$ ) mean the distribution is narrow in that direction.

*Intuition 1.1.* The Gaussian is the maximum-entropy distribution for a given mean and covariance. If all you know about a random variable is its mean and spread, the Gaussian is the “least-assuming” density consistent with those constraints. This is why it appears everywhere as a default model for continuous noise.

## 1.2 Likelihood: turning a distribution into a score

Given a fixed observation  $x_{\text{obs}} \in \mathbb{R}^d$ , we can *evaluate* the density at that point:

$$\ell(\mu, \Sigma) = \mathcal{N}(x_{\text{obs}}; \mu, \Sigma). \quad (2)$$

This number  $\ell$  is called the **likelihood** of the parameters  $(\mu, \Sigma)$  given the data  $x_{\text{obs}}$ . Note:

- The observation  $x_{\text{obs}}$  is **fixed** (it’s the data we collected).
- The parameters  $(\mu, \Sigma)$  are **variable** (we want to find the best ones).
- The likelihood is *not* a probability distribution over  $(\mu, \Sigma)$ —it does not integrate to 1 over parameter space.

With  $P$  independent observations  $x_1, \dots, x_P$ , the joint likelihood is the product:

$$\ell(\mu, \Sigma) = \prod_{i=1}^P \mathcal{N}(x_i; \mu, \Sigma). \quad (3)$$

Taking the logarithm (which preserves the location of the maximum):

$$\log \ell(\mu, \Sigma) = \sum_{i=1}^P \log \mathcal{N}(x_i; \mu, \Sigma) = -\frac{Pd}{2} \log 2\pi - \frac{P}{2} \log |\Sigma| - \frac{1}{2} \sum_i (x_i - \mu)^\top \Sigma^{-1} (x_i - \mu). \quad (4)$$

**Key idea. Maximum likelihood estimation (MLE):** choose  $(\mu, \Sigma)$  to maximize  $\log \ell$ . For a single Gaussian, the MLE is the sample mean  $\hat{\mu} = \frac{1}{P} \sum_i x_i$  and sample covariance  $\hat{\Sigma} = \frac{1}{P} \sum_i (x_i - \hat{\mu})(x_i - \hat{\mu})^\top$ . This is tractable and has a unique closed-form solution.

### 1.3 Conditional distributions

A **conditional distribution**  $p(y | x)$  is a density over  $y$  that depends on a given value of  $x$ . It satisfies  $\int p(y | x) dy = 1$  for each fixed  $x$ . The dependence on  $x$  enters through the parameters of the density.

**Example 1.1** (Linear regression).  $y | x \sim \mathcal{N}(w^\top x, \sigma^2)$ . The mean  $w^\top x$  depends on the input  $x$ ; the variance  $\sigma^2$  does not. For a given  $x$ , this is a proper Gaussian density over  $y$ .

**Example 1.2** (Our residual model). Suppose we have a dynamical system  $\dot{x} = f(x)$  and we have partitioned phase space into  $N$  clusters. Each cluster  $k$  carries a **local model**  $L_k: \mathbb{R}^d \rightarrow \mathbb{R}^d$ —a function that, given any point  $x$ , outputs a *prediction* of the vector field  $f(x)$ . The local model is a deterministic function defined by the cluster’s parameters; it is *not* a random variable.

Concretely,  $L_k$  is an approximation of  $f$  that is built from the cluster’s center  $c_k$  and some local dynamical parameters (a Jacobian matrix, an affine operator, or a Koopman operator—the exact form will be specified in Section 6). For now, all that matters is: **given  $c_k$  and the local parameters,  $L_k(x)$  is a computable function that takes a point  $x \in \mathbb{R}^d$  and returns a vector in  $\mathbb{R}^d$ .** For example, in the simplest case (Taylor linearization):

$$L_k(x) = f(c_k) + \mathbf{J}(c_k)(x - c_k), \quad (5)$$

where  $f(c_k) \in \mathbb{R}^d$  is the vector field evaluated at the center (a known vector) and  $\mathbf{J}(c_k) \in \mathbb{R}^{d \times d}$  is the Jacobian matrix at the center (a known matrix). The computation is: subtract the center, multiply by the Jacobian, add the vector-field value at the center. The result is an affine function of  $x$ .

With  $L_k$  defined, we can specify the conditional distribution:

$$f_{\text{obs}}(x_i) | x_i, z_i = k \sim \mathcal{N}(L_k(x_i), \sigma^2 I). \quad (6)$$

This says: given the input  $x_i$  and the assignment to cluster  $k$ , the observed vector field  $f_{\text{obs}}(x_i)$  is a Gaussian random variable with:

- **Center (“mean”)**  $= L_k(x_i)$ . This is the local model’s prediction of  $f$  at  $x_i$ . It is computed by evaluating the function  $L_k$  at the specific input  $x_i$ —**it is a single deterministic value, not an average over anything**. Different inputs give different values.

We call it “the mean” only because of Gaussian convention: in  $\mathcal{N}(y; \mu, \Sigma)$ , the parameter  $\mu$  is traditionally called “the mean” because if one drew infinitely many samples  $y_1, y_2, \dots$  from this distribution, their sample average would converge to  $\mu$ . But  $\mu$  itself is just a location parameter—the point where the bell curve is centered. In our context, a less ambiguous name is **the predicted value**:  $L_k(x_i)$  is the value that the local model predicts the vector field should take at  $x_i$ , and the Gaussian is centered there.

- **Covariance**  $= \sigma^2 I$ . This is a fixed isotropic matrix that does *not* depend on  $x_i$ . It sets the scale of “tolerable prediction error”—how far the actual observation  $f_{\text{obs}}(x_i)$  can be from the prediction  $L_k(x_i)$  before the likelihood becomes small.

For a given  $x_i$  and  $k$ , this is a proper Gaussian density over  $f_{\text{obs}} \in \mathbb{R}^d$  (integrates to 1 over  $\mathbb{R}^d$ ).

When we evaluate this density at the *observed* value  $f_{\text{obs}}(x_i)$ , we get a **likelihood value**:

$$\mathcal{N}(f_{\text{obs}}(x_i); L_k(x_i), \sigma^2 I) = (2\pi\sigma^2)^{-d/2} \exp\left(-\frac{\|f_{\text{obs}}(x_i) - L_k(x_i)\|^2}{2\sigma^2}\right). \quad (7)$$

This number depends on three things:

1.  $f_{\text{obs}}(x_i)$ : the observed vector field (data—fixed).
2.  $L_k(x_i)$ : the local model’s prediction (computed from current cluster parameters and input  $x_i$ ).
3.  $\sigma^2$ : the noise scale (hyperparameter).

It measures “how well does cluster  $k$ ’s local model predict the vector field at  $x_i$ ?” A small squared error  $\|f_{\text{obs}}(x_i) - L_k(x_i)\|^2$  gives a large likelihood; a large squared error gives a small likelihood.

## 2 Latent Variable Models and the Marginal Likelihood

### 2.1 Why latent variables?

A **mixture model** assumes that each data point  $x_i$  was generated by one of  $N$  components, but we don’t observe which one. The identity of the generating component is a **latent (hidden) variable**  $z_i \in \{1, \dots, N\}$ .

**What is  $x_i$ ?** In our setting,  $x_i \in \mathbb{R}^d$  is a **state vector** of the dynamical system—a single point in phase space. For the Lorenz attractor,  $x_i = (x, y, z) \in \mathbb{R}^3$  is a 3D position on the attractor. For the damped pendulum,  $x_i = (\theta, \dot{\theta}) \in \mathbb{R}^2$  is an angle and angular velocity. We have  $P$  such state vectors, sampled from a trajectory or from the phase space directly.

The generative process for a standard Gaussian mixture:

1. Draw  $z_i \sim \text{Categorical}(\pi_1, \dots, \pi_N)$ , where  $\pi_k \geq 0$ ,  $\sum_k \pi_k = 1$ .
2. Draw  $x_i \sim \mathcal{N}(\mu_{z_i}, \Sigma_{z_i})$ .

Step 2 says: given that state  $x_i$  was assigned to cluster  $k$  (i.e.,  $z_i = k$ ), its location in phase space is drawn from a Gaussian centered at  $\mu_k$  with covariance  $\Sigma_k$ . The term  $p(x_i | z_i = k, \theta) = \mathcal{N}(x_i; \mu_k, \Sigma_k)$  answers the question: “*given that this state belongs to cluster  $k$ , how likely is it to be at this particular location in phase space?*” States close to the cluster center  $\mu_k$  get high probability; states far away get low probability.

We observe  $x_i$  but not  $z_i$ . To write the density of the observed variable  $x_i$  alone, we need to **marginalize out** the latent variable  $z_i$ .

**Definition 2.1** (Marginalization). If two variables  $A$  and  $B$  have a joint distribution  $p(A, B)$ , the **marginal distribution** of  $A$  is obtained by summing (or integrating) over all possible values of  $B$ :

$$p(A) = \sum_{\text{all } b} p(A, B = b). \quad (8)$$

This “removes”  $B$  from the joint by accounting for every possibility.

In our case,  $A = x_i$  (observed) and  $B = z_i$  (latent, takes values  $1, \dots, N$ ). The marginalization formula gives:

$$p(x_i | \theta) = \sum_{k=1}^N p(x_i, z_i = k | \theta). \quad (9)$$

Each term  $p(x_i, z_i = k | \theta)$  is the joint probability of observing  $x_i$  and having it come from component  $k$ . We can decompose this joint using the **chain rule** (also called the product rule) of probability:

$$p(x_i, z_i = k | \theta) = p(z_i = k | \theta) \cdot p(x_i | z_i = k, \theta). \quad (10)$$

This says: the probability of “ $x_i$  was generated by component  $k$ ” equals the probability of “component  $k$  was chosen” times the probability of “component  $k$  generated  $x_i$ .” Now we substitute what we know about each factor:

- $p(z_i = k \mid \theta) = \pi_k$  (by the Categorical prior on  $z_i$ ).
- $p(x_i \mid z_i = k, \theta) = \mathcal{N}(x_i; \mu_k, \Sigma_k)$  (by the Gaussian emission model of component  $k$ ).

Substituting into (10) and then into (9):

$$p(x_i \mid \theta) = \sum_{k=1}^N \pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k). \quad (11)$$

This is the **marginal density** of  $x_i$ —a weighted sum of Gaussians, where  $\theta = (\pi, \{\mu_k, \Sigma_k\}_{k=1}^N)$  collects all parameters. Each term contributes the density of  $x_i$  under component  $k$ , weighted by the prior probability  $\pi_k$  of that component.

## 2.2 Why not just maximize the marginal likelihood?

The log of the marginal likelihood for all data is

$$\log p(X \mid \theta) = \sum_{i=1}^P \log \left( \sum_{k=1}^N \pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k) \right). \quad (12)$$

This expression contains a log of a sum, and we claim it has no closed-form maximizer. Let us see concretely why.

**For a single Gaussian ( $N = 1$ ), there is no problem.** The sum has one term, so

$$\log p(X \mid \theta) = \sum_i \log \mathcal{N}(x_i; \mu, \Sigma).$$

The log applies directly to a single Gaussian—no sum inside. The derivative with respect to  $\mu$  is:

$$\frac{\partial}{\partial \mu} \sum_i \log \mathcal{N}(x_i; \mu, \Sigma) = \sum_i \Sigma^{-1}(x_i - \mu) = 0 \implies \mu = \frac{1}{P} \sum_i x_i.$$

One equation, one unknown, closed-form solution.

**For a mixture ( $N > 1$ ), the parameters are coupled.** Now differentiate with respect to  $\mu_k$  using the chain rule for  $\log(\sum_j \dots)$ :

$$\frac{\partial}{\partial \mu_k} \log \left( \sum_j \pi_j \mathcal{N}(x_i; \mu_j, \Sigma_j) \right) = \frac{\pi_k \Sigma_k^{-1}(x_i - \mu_k) \mathcal{N}(x_i; \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(x_i; \mu_j, \Sigma_j)}. \quad (13)$$

The denominator  $\sum_j \pi_j \mathcal{N}(x_i; \mu_j, \Sigma_j)$  contains **every cluster's parameters**—all  $\mu_j, \Sigma_j, \pi_j$ . Setting the derivative to zero for  $\mu_k$  gives an equation that involves  $\mu_1, \dots, \mu_N, \Sigma_1, \dots, \Sigma_N$ , and  $\pi_1, \dots, \pi_N$  simultaneously. It is a system of coupled nonlinear equations with no closed-form solution.

*Remark 2.1.* The ratio  $\pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k) / \sum_j \pi_j \mathcal{N}(x_i; \mu_j, \Sigma_j)$  is exactly the responsibility  $r_{ik}$ . But here it is a *function of all parameters*, not a fixed number. That is the coupling: you cannot solve for  $\mu_k$  without knowing  $r_{ik}$ , and you cannot compute  $r_{ik}$  without knowing all the  $\mu_j$ . EM breaks this circularity by alternating: fix  $r_{ik}$  (E-step), then solve for  $\mu_k$  (M-step), repeat.

**Key idea.** Latent variable models have intractable marginal likelihoods because the log of a sum couples all cluster parameters together. The EM algorithm breaks this coupling by replacing  $\log(\text{sum})$  with  $\text{sum}(\log)$  (via the ELBO), at the cost of introducing an auxiliary distribution  $q$  that must be iteratively updated.

## 3 The Evidence Lower Bound (ELBO)

### 3.1 The problem: a log of a sum

Our goal is to maximize  $\log p(X | \theta)$  (12) over  $\theta$ . But this expression contains a *logarithm of a sum*, which has no closed-form maximizer. We need a workaround.

The strategy: construct a simpler function—a **lower bound** on  $\log p(X | \theta)$ —that *is* tractable to maximize. If we can make this bound tight (close to the true value), then maximizing the bound approximately maximizes the true objective.

### 3.2 Introducing a helper distribution

The key trick is to introduce an auxiliary distribution  $q$  over the latent variables. Let  $q(z_i)$  be *any* probability distribution over  $\{1, \dots, N\}$ —one for each data point  $i$ . We write  $q(z_i = k)$  for the probability that  $q$  assigns point  $i$  to cluster  $k$ . The only requirements are  $q(z_i = k) \geq 0$  and  $\sum_k q(z_i = k) = 1$ .

*Intuition 3.1.* Think of  $q$  as our current “guess” at the cluster assignments. It does not have to be the true posterior—it can be *anything*. The ELBO derivation works for any  $q$ , and we will later choose  $q$  to make the bound as tight as possible.

### 3.3 Step-by-step derivation

We start from the marginal log-likelihood for a single data point and work through five algebraic steps.

**Step 1: rewrite using marginalization.** The marginal  $p(x_i | \theta)$  is a sum over the latent variable (as derived in Section 2.1):

$$\log p(x_i | \theta) = \log \sum_{k=1}^N p(x_i, z_i = k | \theta). \quad (14)$$

This is just the log of the marginal density. So far, nothing new.

**Step 2: multiply and divide by  $q(z_i = k)$  inside the sum.** For any  $q$  with  $q(z_i = k) > 0$  for all  $k$ :

$$\log \sum_k p(x_i, z_i = k | \theta) = \log \sum_k q(z_i = k) \cdot \frac{p(x_i, z_i = k | \theta)}{q(z_i = k)}. \quad (15)$$

This is a purely algebraic identity—multiplying and dividing by the same positive number changes nothing. **Why do we do this?** Because the sum  $\sum_k q(z_i = k) \cdot (\dots)$  is a **weighted average** (an expectation under  $q$ ), and weighted averages interact nicely with concave functions like log via Jensen’s inequality.

**Step 3: apply Jensen’s inequality to get a lower bound.** Before stating the inequality, let us understand **why we want a lower bound at all**.

Our goal is to find the parameters  $\theta$  that maximize  $\log p(X | \theta)$ . But we cannot maximize  $\log p(X | \theta)$  directly because it contains a log-of-a-sum (equation (14)) that has no closed-form solution.

Suppose instead we had a simpler function  $\mathcal{L}(q, \theta)$  with two properties:

1. **It is a lower bound:**  $\mathcal{L}(q, \theta) \leq \log p(X | \theta)$  for all  $q, \theta$ .

2. **It can be made tight:** there exists a choice of  $q$  such that  $\mathcal{L}(q, \theta) = \log p(X | \theta)$  exactly.

If we have such a function, we can use the following two-step strategy to maximize  $\log p(X | \theta)$ :

- **Tighten the bound** (choose  $q$  to close the gap): after this step,  $\mathcal{L}(q, \theta) = \log p(X | \theta)$ —the bound touches the true objective.
- **Push the bound up** (choose  $\theta$  to increase  $\mathcal{L}$ ): after this step,  $\mathcal{L}$  increases. Since the bound was tight before this step, the true objective  $\log p(X | \theta)$  must also increase (or stay the same).

Alternating these two steps gives a monotonically non-decreasing sequence  $\log p(X | \theta^{(0)}) \leq \log p(X | \theta^{(1)}) \leq \dots$ . **This is the EM algorithm:** the first step is the E-step, the second is the M-step, and  $\mathcal{L}$  is the ELBO.

So: we need a lower bound not for its own sake, but because it gives us a **tractable surrogate** that we can optimize in place of the intractable true objective, with the guarantee that each optimization cycle improves (or maintains) the true objective.

Now we construct such a bound using Jensen’s inequality.

**Definition 3.1** (Jensen’s inequality). If  $\varphi$  is a **concave** function (curves downward, like  $\log$ ) and  $W_1, \dots, W_N$  are non-negative weights summing to 1, then for any values  $a_1, \dots, a_N$ :

$$\varphi\left(\sum_k W_k a_k\right) \geq \sum_k W_k \varphi(a_k). \quad (16)$$

In words: the function of the weighted average is  $\geq$  the weighted average of the function.

We apply (16) to (15) with  $\varphi = \log$  (which is concave),  $W_k = q(z_i = k)$ , and  $a_k = p(x_i, z_i = k | \theta) / q(z_i = k)$ :

$$\log \sum_k q(z_i = k) \cdot \frac{p(x_i, z_i = k | \theta)}{q(z_i = k)} \geq \sum_k q(z_i = k) \cdot \log \frac{p(x_i, z_i = k | \theta)}{q(z_i = k)}. \quad (17)$$

The log has moved **inside** the sum, at the cost of turning an equality into an inequality ( $\geq$ ). The right-hand side is a lower bound on the left-hand side.

**Step 4: split the log of a ratio.** Using  $\log(A/B) = \log A - \log B$ :

$$\begin{aligned} \sum_k q(z_i = k) \log \frac{p(x_i, z_i = k | \theta)}{q(z_i = k)} &= \sum_k q(z_i = k) \log p(x_i, z_i = k | \theta) \\ &\quad - \sum_k q(z_i = k) \log q(z_i = k). \end{aligned} \quad (18)$$

The first sum is the expected joint log-likelihood under  $q$ . The second sum (with the minus sign) is the **entropy** of  $q$ , written  $H[q] = -\sum_k q_k \log q_k$ .

**Step 5: sum over all data points and define the ELBO.** Summing (17)–(18) over  $i = 1, \dots, P$ , and writing  $r_{ik} = q(z_i = k)$ :

$$\log p(X | \theta) \geq \underbrace{\sum_{i,k} r_{ik} \log p(x_i, z_i = k | \theta)}_{\text{expected complete-data log-likelihood}} + \underbrace{\left(-\sum_{i,k} r_{ik} \log r_{ik}\right)}_{\text{entropy } H[q]} \equiv \text{ELBO}(q, \theta). \quad (19)$$



**Key idea.** The ELBO is a **lower bound** on the log marginal likelihood:  $\log p(X | \theta) \geq \text{ELBO}(q, \theta)$  for *any* distribution  $q$  over the latent variables.

This bound holds because Jensen’s inequality introduced a  $\geq$  at Step 3. The bound is tight (equality holds) when  $q(z_i = k) = p(z_i = k | x_i, \theta)$ —i.e., when  $q$  equals the true posterior. In that case the gap is zero:

$$\log p(X | \theta) - \text{ELBO}(q, \theta) = \text{KL}(q(Z) \| p(Z | X, \theta)) \geq 0, \quad (20)$$

where KL is the Kullback-Leibler divergence (a non-negative measure of “distance” between two distributions; it equals zero if and only if the two distributions are identical).

*Intuition 3.2.* Why does this help? Because the ELBO has  $\log p(x_i, z_i = k | \theta)$  (joint) instead of  $\log \sum_k p(x_i, z_i = k | \theta)$  (marginal). The joint factors as  $p(z_i = k) \cdot p(x_i | z_i = k)$  (chain rule), and the log of a product is a sum of logs—no more “log of a sum.” This makes the M-step tractable.

### 3.4 Decomposing the ELBO

To make the ELBO concrete, we need to expand  $\log p(x_i, z_i = k | \theta)$ . Using the chain rule (10):

$$p(x_i, z_i = k | \theta) = p(z_i = k | \theta) \cdot p(x_i | z_i = k, \theta). \quad (21)$$

Substituting our specific distributions (Categorical prior  $\pi_k$ , Gaussian emission):

$$p(x_i, z_i = k | \theta) = \pi_k \cdot \mathcal{N}(x_i; \mu_k, \Sigma_k). \quad (22)$$

Taking the log:  $\log p(x_i, z_i = k | \theta) = \log \pi_k + \log \mathcal{N}(x_i; \mu_k, \Sigma_k)$ . Plugging into the ELBO (19):

$$\text{ELBO}(q, \theta) = \sum_{i,k} r_{ik} \log \pi_k + \sum_{i,k} r_{ik} \log \mathcal{N}(x_i; \mu_k, \Sigma_k) - \sum_{i,k} r_{ik} \log r_{ik}, \quad (23)$$

where we wrote  $r_{ik} = q(z_i = k)$ . The three terms are:

1. **Assignment log-likelihood:**  $\sum_{i,k} r_{ik} \log \pi_k$ . Prefers configurations where points are assigned to clusters with large  $\pi_k$ .
2. **Data log-likelihood:**  $\sum_{i,k} r_{ik} \log \mathcal{N}(x_i; \mu_k, \Sigma_k)$ . Prefers configurations where each point is close to its assigned center.
3. **Entropy:**  $-\sum_{i,k} r_{ik} \log r_{ik}$ . Prefers “soft” (uncertain) assignments. This prevents the trivial solution of putting all mass on one cluster.

## 4 The EM Algorithm

### 4.1 E-step: optimize over $q$

Fix  $\theta$ . We want  $q^* = \arg \max_q \text{ELBO}(q, \theta)$ . Since the gap is the KL divergence  $\text{KL}(q \| p(Z | X, \theta))$ , setting  $q^*(z_i = k) = p(z_i = k | x_i, \theta)$  makes  $\text{KL} = 0$  and the ELBO equals  $\log p(X | \theta)$  exactly.

We compute this using **Bayes’ theorem**.

**Definition 4.1** (Bayes’ theorem). For any two events or variables  $A$  and  $B$  with  $p(B) > 0$ :

$$p(A | B) = \frac{p(B | A) p(A)}{p(B)}. \quad (24)$$

In words: posterior = (likelihood  $\times$  prior) / evidence. The evidence  $p(B)$  is a normalizing constant that ensures the posterior sums/integrates to 1.

We apply (24) with  $A = (z_i = k)$  and  $B = x_i$ :

$$p(z_i = k \mid x_i, \theta) = \frac{p(x_i \mid z_i = k, \theta) \cdot p(z_i = k \mid \theta)}{p(x_i \mid \theta)}. \quad (25)$$

Now we substitute each factor using what we know about our model:

- **Prior:**  $p(z_i = k \mid \theta) = \pi_k$  (Categorical distribution over components).
- **Likelihood:**  $p(x_i \mid z_i = k, \theta) = \mathcal{N}(x_i; \mu_k, \Sigma_k)$  (Gaussian emission of component  $k$ ).
- **Evidence:**  $p(x_i \mid \theta) = \sum_{j=1}^N p(x_i, z_i = j \mid \theta) = \sum_{j=1}^N \pi_j \mathcal{N}(x_i; \mu_j, \Sigma_j)$  (the marginal density, obtained by summing the joint over all components, as derived in Section 2.1).

Substituting into (25):

$$r_{ik} = p(z_i = k \mid x_i, \theta) = \frac{\pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k)}{\sum_{j=1}^N \pi_j \mathcal{N}(x_i; \mu_j, \Sigma_j)}. \quad (26)$$

*Intuition 4.1.* The responsibility  $r_{ik}$  is the posterior probability that point  $i$  belongs to cluster  $k$ . It is large when  $x_i$  is close to  $\mu_k$  (high  $\mathcal{N}$  value) and  $\pi_k$  is large (component is a priori likely). Points equidistant from two centers will have  $r_{ik} \approx r_{ij} \approx 0.5$ —soft assignment.

## 4.2 M-step: optimize over $\theta$

Fix  $q$  (i.e., fix the responsibilities  $r_{ik}$ ). We want  $\theta^* = \arg \max_{\theta} \text{ELBO}(q, \theta)$ . Since the entropy term  $H[q] = -\sum_{i,k} r_{ik} \log r_{ik}$  does not depend on  $\theta$  (the  $r_{ik}$  are fixed numbers), this is equivalent to maximizing the **expected complete-data log-likelihood**:

$$Q(\theta) = \sum_{i,k} r_{ik} \log p(x_i, z_i = k \mid \theta). \quad (27)$$

Now we expand  $\log p(x_i, z_i = k \mid \theta)$  using the chain rule:

$$\begin{aligned} \log p(x_i, z_i = k \mid \theta) &= \log [p(z_i = k \mid \theta) \cdot p(x_i \mid z_i = k, \theta)] \\ &= \log p(z_i = k \mid \theta) + \log p(x_i \mid z_i = k, \theta) \quad (\text{log of a product} = \text{sum of logs}) \\ &= \log \pi_k + \log \mathcal{N}(x_i; \mu_k, \Sigma_k). \quad (\text{substituting our specific distributions}) \end{aligned} \quad (28)$$

Plugging back into (27):

$$Q(\theta) = \sum_{i,k} r_{ik} [\log \pi_k + \log \mathcal{N}(x_i; \mu_k, \Sigma_k)]. \quad (29)$$

**Update for  $\pi_k$ :** Maximize  $\sum_k (\sum_i r_{ik}) \log \pi_k$  subject to  $\sum_k \pi_k = 1$ . By Lagrange multipliers:

$$\pi_k^{\text{new}} = \frac{R_k}{P}, \quad R_k = \sum_{i=1}^P r_{ik}. \quad (30)$$

$R_k$  is the **effective mass** of cluster  $k$ —the total soft-assignment weight. The updated  $\pi_k$  is just the fraction of total mass in cluster  $k$ .

**Update for  $\mu_k$ :** Maximize  $\sum_i r_{ik} \log \mathcal{N}(x_i; \mu_k, \Sigma_k)$  over  $\mu_k$ . The log-Gaussian contains  $-\frac{1}{2}(x_i - \mu_k)^\top \Sigma_k^{-1}(x_i - \mu_k)$ . Taking the gradient:

$$\nabla_{\mu_k} Q = \sum_i r_{ik} \Sigma_k^{-1}(x_i - \mu_k) = 0 \implies \mu_k^{\text{new}} = \frac{\sum_i r_{ik} x_i}{R_k}. \quad (31)$$

The new center is the **responsibility-weighted mean** of the data.

**Update for  $\Sigma_k$ :** Maximize over  $\Sigma_k$ . This involves the log-determinant and the quadratic form. Taking the matrix derivative and setting to zero:

$$\Sigma_k^{\text{new}} = \frac{1}{R_k} \sum_i r_{ik} (x_i - \mu_k^{\text{new}})(x_i - \mu_k^{\text{new}})^\top. \quad (32)$$

The new covariance is the **responsibility-weighted scatter** around the new mean.

**Key idea. EM monotonicity:** Each E-step makes  $\text{ELBO}(q, \theta) = \log p(X | \theta)$  (closes the KL gap). Each M-step maximizes  $Q(\theta)$  over  $\theta$ , which increases ELBO (and hence  $\log p(X | \theta)$ ). The sequence of marginal likelihoods is non-decreasing:

$$\log p(X | \theta^{(0)}) \leq \log p(X | \theta^{(1)}) \leq \log p(X | \theta^{(2)}) \leq \dots$$

Convergence to a local maximum is guaranteed.

## 5 Adding Bayesian Priors

### 5.1 Why priors?

MLE can overfit: with  $N$  clusters and  $P$  points, if a cluster contains very few points, its covariance estimate  $\hat{\Sigma}_k$  becomes singular (not invertible). Bayesian priors **regularize** by encoding prior beliefs about parameter ranges. The ELBO becomes:

$$\text{ELBO}(q, \theta) = \mathbb{E}_q[\log p(X, Z | \theta)] + H[q] + \log p(\theta), \quad (33)$$

where  $\log p(\theta)$  is the log-prior on all parameters. The M-step now maximizes  $Q(\theta) + \log p(\theta)$ , which is the **MAP (maximum a posteriori)** objective.

### 5.2 The Dirichlet prior on mixing weights

The mixing weights  $\pi = (\pi_1, \dots, \pi_N)$  live on the probability simplex ( $\pi_k \geq 0$ ,  $\sum_k \pi_k = 1$ ). The natural prior is the **Dirichlet distribution**:

$$p(\pi | \alpha_0) = \frac{\Gamma(N\alpha_0)}{\Gamma(\alpha_0)^N} \prod_{k=1}^N \pi_k^{\alpha_0-1}. \quad (34)$$

The parameter  $\alpha_0 > 0$  controls the prior's character:

- $\alpha_0 = 1$ : uniform prior on the simplex (all weight allocations equally likely).
- $\alpha_0 > 1$ : prefers  $\pi$  near uniform ( $\pi_k \approx 1/N$  for all  $k$ ). Discourages empty clusters.
- $\alpha_0 < 1$ : **sparse-inducing**. Prefers  $\pi$  concentrated on a few components, with many  $\pi_k \approx 0$ .

**Key idea.** We use  $\alpha_0 = 0.5 < 1$  to encourage the model to **prune unused clusters**. Starting with  $N$  larger than needed, the Dirichlet prior drives  $\pi_k \rightarrow 0$  for redundant components. This is automatic model-complexity selection.

**MAP update for  $\pi$ :** Adding  $\log p(\pi) = \sum_k (\alpha_0 - 1) \log \pi_k + \text{const}$  to the Lagrangian:

$$\pi_k^{\text{new}} = \frac{R_k + \alpha_0 - 1}{P + N(\alpha_0 - 1)}. \quad (35)$$

When  $\alpha_0 < 1$  and  $R_k$  is small,  $\pi_k$  can go negative; we clamp to  $\pi_k \geq 10^{-10}$  and renormalize. In practice, clusters with  $R_k < 1$  are pruned (removed from the model entirely).

### 5.3 The Normal-Inverse-Wishart (NIW) prior on centers and covariances

For the center  $c_k$  and covariance  $\Sigma_k$  of each cluster, we use the **NIW prior**:

$$\Sigma_k \sim \mathcal{IW}(\Psi_0, \nu_0), \quad (36)$$

$$c_k \mid \Sigma_k \sim \mathcal{N}(\mu_0, \Sigma_k / \kappa_0), \quad (37)$$

where  $\Psi_0 \in \mathbb{R}^{d \times d}$  (positive definite scale matrix),  $\nu_0 > d - 1$  (degrees of freedom),  $\mu_0 \in \mathbb{R}^d$  (prior mean),  $\kappa_0 > 0$  (prior precision scaling).

*Intuition 5.1.* The NIW prior encodes:

- $\mu_0$ : “we expect cluster centers near this point” (set to the data mean).
- $\kappa_0$ : “how strongly we believe that” (small  $\kappa_0$  = weak prior = centers free to move; large  $\kappa_0$  = strong pull toward  $\mu_0$ ).
- $\Psi_0$ : “we expect covariances of roughly this scale” (set proportional to identity).
- $\nu_0$ : “how confident we are about the covariance” (small  $\nu_0$  = vague prior; large = confident).

**Why “conjugate”?** A prior is **conjugate** to a likelihood if the posterior has the same functional form as the prior. For a Gaussian likelihood with NIW prior, the posterior on  $(c_k, \Sigma_k)$  is also NIW with updated parameters:

$$\kappa_n = \kappa_0 + R_k, \quad \nu_n = \nu_0 + R_k, \quad (38)$$

$$\mu_n = \frac{\kappa_0 \mu_0 + R_k \bar{x}_k}{\kappa_n}, \quad (39)$$

$$\Psi_n = \Psi_0 + S_k + \frac{\kappa_0 R_k}{\kappa_n} (\bar{x}_k - \mu_0)(\bar{x}_k - \mu_0)^\top, \quad (40)$$

where  $\bar{x}_k = \frac{1}{R_k} \sum_i r_{ik} x_i$  is the responsibility-weighted mean and  $S_k = \sum_i r_{ik} (x_i - \bar{x}_k)(x_i - \bar{x}_k)^\top$  is the weighted scatter.

Conjugacy is important because it gives **closed-form marginal likelihoods** (Section 9).

**MAP updates for  $c_k$  and  $\Sigma_k$ :** Setting the gradient of  $Q(\theta) + \log p(\theta)$  to zero for  $c_k$ :

$$(\kappa_0 \Sigma_k^{-1} + R_k \Sigma_k^{-1}) c_k^{\text{new}} = \kappa_0 \Sigma_k^{-1} \mu_0 + \Sigma_k^{-1} \sum_i r_{ik} x_i. \quad (41)$$

Since  $\Sigma_k^{-1}$  factors out (and we often write  $\Lambda_0 = \kappa_0 \Sigma_k^{-1}$  or a fixed matrix):

$$c_k^{\text{new}} = \frac{\kappa_0 \mu_0 + \sum_i r_{ik} x_i}{\kappa_0 + R_k} = \frac{\kappa_0 \mu_0 + R_k \bar{x}_k}{\kappa_n}. \quad (42)$$

This is a **weighted average** of the prior mean  $\mu_0$  and the data mean  $\bar{x}_k$ , with weights  $\kappa_0$  (prior) and  $R_k$  (data). More data  $\rightarrow$  center follows data; less data  $\rightarrow$  center stays near  $\mu_0$ .

For  $\Sigma_k$ , the MAP mode of the Inverse-Wishart posterior is:

$$\Sigma_k^{\text{new}} = \frac{\Psi_n}{\nu_n + d + 1} = \frac{\Psi_0 + S_k + \text{conflict}}{\nu_0 + R_k + d + 1}. \quad (43)$$

The “conflict” term  $\frac{\kappa_0 R_k}{\kappa_n} (\bar{x}_k - \mu_0)(\bar{x}_k - \mu_0)^\top$  penalizes disagreement between the prior mean and data mean. In our implementation, we use a simplified version where  $\Lambda_0$  is a fixed precision matrix:

$$\Sigma_k^{\text{new}} = \frac{\Psi_0 + \sum_i r_{ik} (x_i - c_k^{\text{new}})(x_i - c_k^{\text{new}})^\top}{\nu_0 + R_k + d + 1} + \lambda I. \quad (44)$$

The  $\lambda I$  term (typically  $10^{-6}$ ) prevents numerical singularity.

## 6 Our Model: Adding the Residual Factor

**Key idea. Connection to Cluster-Weighted Models.** The model we are about to define is an instance of the **Cluster-Weighted Model** (CWM), a framework introduced by Gershenfeld (1997) and formalized statistically by Ingrassia, Minotti, and Vittadini (2012). A CWM models the joint density  $p(x, y) = \sum_k \pi_k p(x|k) p(y|x, k)$ —combining a marginal density on inputs  $p(x|k)$  with a conditional regression on outputs  $p(y|x, k)$ . In the Linear Gaussian CWM, both are Gaussian with an affine conditional mean.

Our model is a CWM where  $x$  is a state vector and  $y = f(x)$  is the vector field. The standard data-driven affine local model (our Variant B) is exactly a Linear Gaussian CWM. Our contributions are: (i) a physics-constrained variant where the local model is the analytic Taylor expansion (Variant A, no precedent in CWM literature), (ii) a Koopman-theoretic variant with lifted-space generators (Variant C), and (iii) application to dynamical-systems phase-space partitioning, an entirely new domain for CWM.

Sections 1–5 presented standard Bayesian GMM machinery. From this point on, everything is specific to the CWM formulation for dynamical systems.

### 6.1 The generative model

We augment the standard GMM with a **second observation per data point**: the vector-field value  $f(x_i)$ . The generative story for each point:

$$z_i \sim \text{Cat}(\pi), \quad (45)$$

$$x_i \mid z_i = k \sim \mathcal{N}(c_k, \Sigma_k), \quad (46)$$

$$f_{\text{obs}}(x_i) \mid x_i, z_i = k \sim \mathcal{N}(L_k(x_i), \sigma^2 I). \quad (47)$$

Line (46) is the proximity model (standard GMM). Line (47) is the residual model: it says that the observed vector field  $f_{\text{obs}}(x_i)$  is drawn from a Gaussian centered at the local model’s prediction  $L_k(x_i)$ , with isotropic noise  $\sigma^2 I$ . Together, lines (46)–(47) define a **Cluster-Weighted Model** (CWM)—the joint-density factorization  $p(x, y) = \sum_k \pi_k p(x|k) p(y|x, k)$  introduced by Gershenfeld (1997) and formalized by Ingrassia et al. (2012).

**What is  $L_k(x_i)$ ?** This is the local model’s prediction of the vector field at  $x_i$ . It depends on:

- The cluster’s center  $c_k$  and local-model parameters (e.g., Jacobian  $\mathbf{J}_k$ , or Koopman operator  $\mathbf{M}_k$ ).
- The input point  $x_i$  itself.

Its specific form depends on the variant (Taylor, LS, or EDMD). The framework is agnostic to this choice.

## 6.2 The joint observation likelihood

For a single data point  $(x_i, f_{\text{obs}}(x_i))$  assigned to cluster  $k$ , we want the joint density  $p(x_i, f_{\text{obs}}(x_i) \mid z_i = k, \theta)$ . We build it in three steps.

**Step 1: apply the chain rule of probability.** For any two variables  $A$  and  $B$ :

$$p(A, B) = p(A) \cdot p(B \mid A). \quad (48)$$

This decomposes a joint density into a marginal times a conditional. Applying this with  $A = x_i$  and  $B = f_{\text{obs}}(x_i)$ , all conditioned on  $z_i = k$ :

$$p(x_i, f_{\text{obs}}(x_i) \mid z_i = k, \theta) = \underbrace{p(x_i \mid z_i = k, \theta)}_{\text{factor 1}} \cdot \underbrace{p(f_{\text{obs}}(x_i) \mid x_i, z_i = k, \theta)}_{\text{factor 2}}. \quad (49)$$

**Step 2: identify each factor from our generative model.**

- **Factor 1** (proximity): From (46), the generative model says  $x_i \mid z_i = k \sim \mathcal{N}(c_k, \Sigma_k)$ . So:

$$p(x_i \mid z_i = k, \theta) = \mathcal{N}(x_i; c_k, \Sigma_k). \quad (50)$$

- **Factor 2** (residual): From (47), the generative model says  $f_{\text{obs}}(x_i) \mid x_i, z_i = k \sim \mathcal{N}(L_k(x_i), \sigma^2 I)$ . So:

$$p(f_{\text{obs}}(x_i) \mid x_i, z_i = k, \theta) = \mathcal{N}(f_{\text{obs}}(x_i); L_k(x_i), \sigma^2 I). \quad (51)$$

**Step 3: substitute into the chain rule (49).**

$$p(x_i, f_{\text{obs}}(x_i) \mid z_i = k, \theta) = \underbrace{\mathcal{N}(x_i; c_k, \Sigma_k)}_{\text{proximity}} \cdot \underbrace{\mathcal{N}(f_{\text{obs}}(x_i); L_k(x_i), \sigma^2 I)}_{\text{residual}}. \quad (52)$$

This is a product of two Gaussian densities over **different random variables**: the first is a density over  $x_i \in \mathbb{R}^d$ , the second is a density over  $f_{\text{obs}}(x_i) \in \mathbb{R}^d$ . Together they define a density over the pair  $(x_i, f_{\text{obs}}(x_i)) \in \mathbb{R}^{2d}$ .

**Derivation. Why is (48) valid here?** In the generative model,  $x_i$  is drawn first (from the proximity model), then  $f_{\text{obs}}(x_i)$  is drawn conditionally on  $x_i$  (from the residual model). The chain rule of probability is a universal identity—it holds for any joint distribution, not just ours. We are simply choosing to decompose  $p(x_i, f_{\text{obs}})$  into  $p(x_i) \cdot p(f_{\text{obs}} \mid x_i)$  rather than the other way around, because this matches the order of our generative process.

### 6.3 The log-likelihood of the joint observation

Taking the log:

$$\begin{aligned}
& \log p(x_i, f_{\text{obs}}(x_i) \mid z_i = k) \\
&= \underbrace{-\frac{d}{2} \log 2\pi - \frac{1}{2} \log |\Sigma_k| - \frac{1}{2} (x_i - c_k)^\top \Sigma_k^{-1} (x_i - c_k)}_{\text{proximity: depends on } x_i, c_k, \Sigma_k} \\
&+ \underbrace{-\frac{d}{2} \log(2\pi\sigma^2) - \frac{\|f_{\text{obs}}(x_i) - L_k(x_i)\|^2}{2\sigma^2}}_{\text{residual: depends on } f_{\text{obs}}(x_i), L_k(x_i), \sigma^2}.
\end{aligned} \tag{53}$$

**Key idea.** The log-likelihood is the **sum of two quadratic penalties**:

1. **Proximity penalty:**  $(x_i - c_k)^\top \Sigma_k^{-1} (x_i - c_k)$  — how far is  $x_i$  from the center in the  $\Sigma_k$ -metric?
2. **Residual penalty:**  $\|f_{\text{obs}}(x_i) - L_k(x_i)\|^2 / \sigma^2$  — how badly does the local model predict  $f$  at this point?

The relative importance of these two penalties is controlled by the ratio of their scales:  $\lambda(\Sigma_k^{-1})$  vs.  $1/\sigma^2$ . When  $\sigma^2$  is small, the residual penalty dominates; when  $\sigma^2$  is large, the proximity penalty dominates.

### 6.4 The role of the residual scale

$\sigma^2$  is the single most important hyperparameter in the framework. It controls the **balance between the two forces**:

- $\sigma^2 \rightarrow \infty$ : The residual penalty  $\|\varepsilon_k\|^2 / (2\sigma^2) \rightarrow 0$ . The joint likelihood reduces to the proximity factor alone. **You recover standard GMM.** Clusters form purely by geometry, ignoring dynamics.
- $\sigma^2 \rightarrow 0$ : The residual penalty dominates everything. A point is assigned to whichever cluster has the smallest  $\|\varepsilon_k\|$ , regardless of geometric proximity. **You get a piecewise regression** where clusters form by model fit alone.
- $\sigma^2 \sim \text{median}(\|\varepsilon_k\|^2)/d$ : Both forces contribute comparably. **This is the interesting regime.** Clusters form by a combination of geometric proximity and model quality.

**Auto-calibration:** We set  $\sigma^2$  from the initial GMM partition's residuals:

$$\sigma^2 = \max\left(\frac{\text{median}_i \|\varepsilon_k(x_i)\|^2}{d}, 10^{-3}\right). \tag{54}$$

This ensures the residual penalty is on the same scale as the actual linearization errors in the initial (geometric) partition. The floor  $10^{-3}$  prevents numerical instability when residuals are very small (e.g., on polynomial systems where the local model is exact).

**Could  $\sigma^2$  be learned?** Yes. The M-step update would be:

$$\sigma^{2,\text{new}} = \frac{\sum_{i,k} r_{ik} \|\varepsilon_k(x_i)\|^2}{d \cdot P}. \quad (55)$$

But this can collapse to  $\sigma^2 \rightarrow \infty$  (residual becomes irrelevant) or  $\sigma^2 \rightarrow 0$  (proximity becomes irrelevant), both of which are local optima of the ELBO. A prior  $\sigma^2 \sim \text{InverseGamma}(a_0, b_0)$  would prevent collapse but adds two more hyperparameters. We defer this and use the auto-calibrated fixed value.

## 7 E-step for Our Model

The E-step computes the posterior  $p(z_i = k \mid x_i, f_{\text{obs}}(x_i), \theta)$ —the probability that point  $i$  belongs to cluster  $k$ , given *both* the position  $x_i$  and the observed vector field  $f_{\text{obs}}(x_i)$ . We derive this in three steps.

**Step 1: apply Bayes’ theorem (24).** Set  $A = (z_i = k)$  and  $B = (x_i, f_{\text{obs}}(x_i))$  (i.e., the “data” is now the *pair* of observations):

$$p(z_i = k \mid x_i, f_{\text{obs}}(x_i), \theta) = \frac{p(x_i, f_{\text{obs}}(x_i) \mid z_i = k, \theta) \cdot p(z_i = k \mid \theta)}{p(x_i, f_{\text{obs}}(x_i) \mid \theta)}. \quad (56)$$

**Step 2: identify each factor from our model.**

- **Likelihood:**  $p(x_i, f_{\text{obs}}(x_i) \mid z_i = k, \theta)$  is the joint observation likelihood (52), which we already derived:  $= \mathcal{N}(x_i; c_k, \Sigma_k) \cdot \mathcal{N}(f_{\text{obs}}(x_i); L_k(x_i), \sigma^2 I)$ .
- **Prior:**  $p(z_i = k \mid \theta) = \pi_k$  (Categorical).
- **Evidence:**  $p(x_i, f_{\text{obs}}(x_i) \mid \theta) = \sum_{j=1}^N p(x_i, f_{\text{obs}}(x_i), z_i = j \mid \theta)$  (marginalize over all clusters, same pattern as Section 2.1). Using the chain rule on each term:  $= \sum_{j=1}^N \pi_j \cdot \mathcal{N}(x_i; c_j, \Sigma_j) \cdot \mathcal{N}(f_{\text{obs}}(x_i); L_j(x_i), \sigma^2 I)$ .

**Step 3: substitute into (56).** Writing  $\varepsilon_k(x_i) = f_{\text{obs}}(x_i) - L_k(x_i)$ :

$$r_{ik} = p(z_i = k \mid x_i, f_{\text{obs}}(x_i), \theta) = \frac{\pi_k \mathcal{N}(x_i; c_k, \Sigma_k) \mathcal{N}(\varepsilon_k(x_i); 0, \sigma^2 I)}{\sum_{j=1}^N \pi_j \mathcal{N}(x_i; c_j, \Sigma_j) \mathcal{N}(\varepsilon_j(x_i); 0, \sigma^2 I)}. \quad (57)$$

Compare to standard GMM (26): the only difference is the extra factor  $\mathcal{N}(\varepsilon_k(x_i); 0, \sigma^2 I)$  in both numerator and denominator. This factor **upweights** clusters where the residual is small (good local fit) and **downweights** clusters where the residual is large (bad local fit).

**Numerical implementation:** We work in log-space to avoid underflow:

$$\log \tilde{r}_{ik} = \log \pi_k + \log \mathcal{N}(x_i; c_k, \Sigma_k) + \log \mathcal{N}(\varepsilon_k(x_i); 0, \sigma^2 I), \quad (58)$$

$$\log r_{ik} = \log \tilde{r}_{ik} - \underset{k}{\text{logsumexp}} \log \tilde{r}_{ik}, \quad (59)$$

where  $\text{logsumexp}_k(a_k) = \log \sum_k \exp(a_k)$ , computed via the max-shift trick for numerical stability.



## 8 M-step for Our Model

The M-step maximizes the expected complete-data log-likelihood  $Q(\theta)$  plus log-priors over  $\theta$ . We build  $Q$  step by step.

The general form of  $Q$  is (27):  $Q(\theta) = \sum_{i,k} r_{ik} \log p(x_i, f_{\text{obs}}(x_i), z_i = k \mid \theta)$ . Using the chain rule to decompose the joint:

$$\begin{aligned} \log p(x_i, f_{\text{obs}}(x_i), z_i = k \mid \theta) &= \log p(z_i = k \mid \theta) + \log p(x_i \mid z_i = k, \theta) + \log p(f_{\text{obs}}(x_i) \mid x_i, z_i = k, \theta) \\ &= \log \pi_k + \log \mathcal{N}(x_i; c_k, \Sigma_k) + \log \mathcal{N}(\varepsilon_k(x_i); 0, \sigma^2 I). \end{aligned} \quad (60)$$

The first two terms are identical to standard GMM. The third term—the log-residual—is what distinguishes the CWM from a standard GMM: it couples the cluster assignments to the local regression quality. This factorization is due to Gershensfeld (1997) and Ingrassia et al. (2012). Substituting into  $Q$ :

$$Q(\theta) = \underbrace{\sum_{i,k} r_{ik} \log \pi_k}_{Q_\pi} + \underbrace{\sum_{i,k} r_{ik} \log \mathcal{N}(x_i; c_k, \Sigma_k)}_{Q_{\text{prox}}} + \underbrace{\sum_{i,k} r_{ik} \log \mathcal{N}(\varepsilon_k(x_i); 0, \sigma^2 I)}_{Q_{\text{res}}}. \quad (61)$$

The  $Q_\pi$  and  $Q_{\text{prox}}$  terms are identical to standard GMM. The  $Q_{\text{res}}$  term (the CWM regression component) is:

$$Q_{\text{res}} = -\frac{Pd}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i,k} r_{ik} \|\varepsilon_k(x_i)\|^2. \quad (62)$$

The  $\pi$  and  $\Sigma_k$  updates are the same as in Sections 4–5. What changes is the **center update**:  $Q_{\text{res}}$  depends on  $c_k$  through  $\varepsilon_k(x_i) = f(x_i) - L_k(x_i)$ , and  $L_k$  depends on  $c_k$ . How this dependence is handled defines the three variants.

### 8.1 Variant A: Taylor-analytic

Here  $L_k(x) = f(c_k) + \mathbf{J}(c_k)(x - c_k)$ , where  $f$  and  $\mathbf{J}$  are evaluated analytically. The center  $c_k$  enters  $Q_{\text{res}}$  through both  $f(c_k)$  and  $\mathbf{J}(c_k)$ . In the M-step, we:

1. Update  $c_k$  using only the proximity gradient:

$$(\Lambda_0 + R_k \Sigma_k^{-1}) c_k^{\text{new}} = \Lambda_0 \mu_0 + \Sigma_k^{-1} \sum_i r_{ik} x_i. \quad (63)$$

This is the standard GMM center update with NIW prior.

2. **Re-tie**: set  $f(c_k) \leftarrow f(c_k^{\text{new}})$ ,  $\mathbf{J}(c_k) \leftarrow \mathbf{J}(c_k^{\text{new}})$  analytically.

*Remark 8.1* (Why this is approximate). The gradient  $\nabla_{c_k} Q_{\text{res}}$  is nonzero—it contains a term from  $\partial \varepsilon_k / \partial c_k$ . By ignoring it in step 1, we are not fully maximizing  $Q$  over  $c_k$ . This makes the update a **generalized EM (GEM)** step: it increases  $Q$  but does not maximize it. The ELBO is still non-decreasing at each step (GEM guarantees this), but it may not converge to the same point as full maximization.

Additionally, step 2 “re-ties”  $f(c_k)$  and  $\mathbf{J}(c_k)$  to the analytic values at the new center. This changes the residual in a way that is *not* accounted for by the gradient—it’s a separate heuristic update. In practice, at small  $N$  (large clusters, large  $\sigma^2$ ), this is harmless. At large  $N$  (small clusters, small  $\sigma^2$ ), the residual term dominates and the non-variational re-tying can decrease the ELBO.

## 8.2 Variant B: Taylor-LS

Here  $L_k(x) = f_k + \mathbf{J}_k(x - c_k)$ , where  $(f_k, \mathbf{J}_k)$  are **free parameters** fit by weighted least squares. We now have three parameter groups to update:  $(f_k, \mathbf{J}_k)$ ,  $c_k$ , and  $\Sigma_k$ . We update them sequentially (coordinate ascent):

**Step 1: update  $(f_k, \mathbf{J}_k)$  by weighted LS.** The residual is  $\varepsilon_k(x_i) = f_{\text{obs}}(x_i) - f_k - \mathbf{J}_k(x_i - c_k)$ . Fix  $c_k$ . The  $Q_{\text{res}}$  term is:

$$Q_{\text{res}}(f_k, \mathbf{J}_k) = -\frac{1}{2\sigma^2} \sum_i r_{ik} \|f_{\text{obs}}(x_i) - f_k - \mathbf{J}_k(x_i - c_k)\|^2 + \text{const.} \quad (64)$$

This is a **weighted least-squares** problem. Define:

$$\phi_i = \begin{bmatrix} 1 \\ x_i - c_k \end{bmatrix} \in \mathbb{R}^{d+1}, \quad Z = \begin{bmatrix} \phi_1^\top \\ \vdots \\ \phi_P^\top \end{bmatrix} \in \mathbb{R}^{P \times (d+1)}, \quad (65)$$

$$W = \text{diag}(r_{1k}, \dots, r_{Pk}), \quad Y = \begin{bmatrix} f_{\text{obs}}(x_1)^\top \\ \vdots \\ f_{\text{obs}}(x_P)^\top \end{bmatrix} \in \mathbb{R}^{P \times d}. \quad (66)$$

The optimal  $B = [f_k^\top; \mathbf{J}_k^\top] \in \mathbb{R}^{(d+1) \times d}$  satisfies the normal equations:

$$(Z^\top W Z + \lambda I) B = Z^\top W Y. \quad (67)$$

The ridge  $\lambda I$  (typically  $\lambda = 10^{-6}$ ) regularizes against ill-conditioning when  $R_k$  is small.

**Derivation.** Take  $\nabla_{f_k} Q_{\text{res}} = 0$ :

$$\sum_i r_{ik} (f_{\text{obs}}(x_i) - f_k - \mathbf{J}_k(x_i - c_k)) = 0 \implies R_k f_k = \sum_i r_{ik} f_{\text{obs}}(x_i) - \mathbf{J}_k \sum_i r_{ik} (x_i - c_k).$$

This, together with the analogous equation for  $\mathbf{J}_k$ , is equivalent to the normal equations (67).

**Step 2: update  $c_k$  (full gradient, including residual).** Now fix the new  $(f_k, \mathbf{J}_k)$  and optimize  $c_k$ . The  $Q$ -function depends on  $c_k$  through both the proximity term and the residual term. We need:

$$\nabla_{c_k} Q = \nabla_{c_k} Q_{\text{prox}} + \nabla_{c_k} Q_{\text{res}} + \nabla_{c_k} \log p(c_k) = 0. \quad (68)$$

**Derivation. Proximity gradient:**  $Q_{\text{prox}}$  contains  $-\frac{1}{2} \sum_i r_{ik} (x_i - c_k)^\top \Sigma_k^{-1} (x_i - c_k)$ . Taking the gradient:

$$\nabla_{c_k} Q_{\text{prox}} = \Sigma_k^{-1} \sum_i r_{ik} (x_i - c_k) = \Sigma_k^{-1} (S_x - R_k c_k),$$

where  $S_x = \sum_i r_{ik} x_i$ .

**Residual gradient:**  $Q_{\text{res}}$  contains  $-\frac{1}{2\sigma^2} \sum_i r_{ik} \|\varepsilon_k(x_i)\|^2$  with  $\varepsilon_k(x_i) = f(x_i) - f_k - \mathbf{J}_k(x_i - c_k)$ . Note  $\frac{\partial \varepsilon_k}{\partial c_k} = \mathbf{J}_k$  (since  $\frac{\partial}{\partial c_k} [-\mathbf{J}_k(x_i - c_k)] = \mathbf{J}_k$ ). So:

$$\begin{aligned} \nabla_{c_k} Q_{\text{res}} &= -\frac{1}{\sigma^2} \sum_i r_{ik} \mathbf{J}_k^\top \varepsilon_k(x_i) \\ &= -\frac{1}{\sigma^2} \mathbf{J}_k^\top [S_f - R_k f_k - \mathbf{J}_k (S_x - R_k c_k)], \end{aligned} \quad (69)$$

where  $S_f = \sum_i r_{ik} f_{\text{obs}}(x_i)$ .

**Prior gradient:**  $\log p(c_k) \supset -\frac{1}{2}(c_k - \mu_0)^\top \Lambda_0 (c_k - \mu_0)$ , so  $\nabla_{c_k} \log p(c_k) = \Lambda_0(\mu_0 - c_k)$ .

**Setting the total gradient to zero:**

$$\Lambda_0(\mu_0 - c_k) + \Sigma_k^{-1}(S_x - R_k c_k) - \frac{1}{\sigma^2} \mathbf{J}_k^\top [S_f - R_k f_k - \mathbf{J}_k(S_x - R_k c_k)] = 0. \quad (70)$$

Collecting all terms proportional to  $c_k$ :

$$\underbrace{\left[ \Lambda_0 + R_k \Sigma_k^{-1} + \frac{R_k}{\sigma^2} \mathbf{J}_k^\top \mathbf{J}_k \right]}_{\text{LHS}} c_k^{\text{new}} = \underbrace{\Lambda_0 \mu_0 + \Sigma_k^{-1} S_x - \frac{1}{\sigma^2} \mathbf{J}_k^\top (S_f - R_k f_k - \mathbf{J}_k S_x)}_{\text{RHS}}. \quad (71)$$

**Key idea.** Equation (71) has three contributions to the left-hand side (the “effective precision” of the center):

1.  $\Lambda_0$ : pull toward the prior mean  $\mu_0$ .
2.  $R_k \Sigma_k^{-1}$ : pull toward the data mean  $S_x/R_k$  (standard GMM term).
3.  $(R_k/\sigma^2) \mathbf{J}_k^\top \mathbf{J}_k$ : pull toward the point that minimizes the residual (the CWM regression term).

**Variant A (Taylor-analytic) drops the third term entirely.** This is what makes it approximate.

**Step 3: update  $\Sigma_k$ .** Same as standard GMM with NIW posterior (no residual dependence on  $\Sigma_k$ ):

$$\Sigma_k^{\text{new}} = \frac{\Psi_0 + \sum_i r_{ik} (x_i - c_k^{\text{new}})(x_i - c_k^{\text{new}})^\top}{\nu_0 + R_k + d + 1} + \lambda I. \quad (72)$$

### 8.3 Variant C: Local EDMD

Here  $L_k(x) = [\mathbf{M}_k \Phi(x - c_k)]_{1:d}$ , where  $\Phi: \mathbb{R}^d \rightarrow \mathbb{R}^M$  is a polynomial lifting and  $\mathbf{M}_k \in \mathbb{R}^{M \times M}$  is the local Koopman generator.

**What is the polynomial lifting?** For degree  $p$  in  $d$  dimensions,  $\Phi(u)$  is the vector of all monomials  $u_1^{a_1} \cdots u_d^{a_d}$  with  $\sum_j a_j \leq p$ . For example, with  $d = 2, p = 2$ :

$$\Phi(u_1, u_2) = (1, u_1, u_2, u_1^2, u_1 u_2, u_2^2)^\top \in \mathbb{R}^6. \quad (73)$$

The total number of monomials is  $M = \binom{d+p}{p}$ .

**Why  $[\mathbf{M}_k \Phi]_{1:d}$  gives  $\hat{f}$ :** The first entry of  $\Phi$  is the constant 1; entries 2 through  $d+1$  are the linear monomials  $u_1, \dots, u_d$ . Their time derivatives are  $\dot{u}_j = \dot{x}_j = f_j(x)$ . The Koopman generator  $\mathbf{M}_k$  is fit so that  $\mathbf{M}_k \Phi(u) \approx \dot{\Phi}(u)$ , where  $\dot{\Phi}$  is the vector of time derivatives of all monomials. Entries 2 through  $d+1$  of  $\dot{\Phi}$  are exactly  $f_1(x), \dots, f_d(x)$ . So:

$$[\mathbf{M}_k \Phi(x - c_k)]_j \approx \dot{\Phi}_j = f_j(x), \quad j = 1, \dots, d. \quad (74)$$

**M-step for  $M_k$ :** The generator is fit by weighted least squares. The target for each data point is  $\dot{\Phi}(x_i - c_k) = \nabla \Phi(x_i - c_k) \cdot f(x_i)$ , the Lie derivative of  $\Phi$  along  $f$ . The feature is  $\Phi(x_i - c_k)$ . With weights  $r_{ik}$ :

$$G_k = \sum_i r_{ik} \Phi(x_i - c_k) \Phi(x_i - c_k)^\top \in \mathbb{R}^{M \times M}, \quad (75)$$

$$A_k = \sum_i r_{ik} \dot{\Phi}(x_i - c_k) \Phi(x_i - c_k)^\top \in \mathbb{R}^{M \times M}, \quad (76)$$

$$M_k = A_k(G_k + \lambda I)^{-1}. \quad (77)$$

**Derivation.** The objective is  $\min_{M_k} \sum_i r_{ik} \|\dot{\Phi}_i - M_k \Phi_i\|^2$ . This is a weighted linear regression in  $\mathbb{R}^M$ : the matrix  $M_k$  is the “weight matrix.” Taking the matrix derivative:

$$\nabla_{M_k} = -2 \sum_i r_{ik} (\dot{\Phi}_i - M_k \Phi_i) \Phi_i^\top = 0 \implies M_k \sum_i r_{ik} \Phi_i \Phi_i^\top = \sum_i r_{ik} \dot{\Phi}_i \Phi_i^\top.$$

This gives  $M_k G_k = A_k$ , hence  $M_k = A_k G_k^{-1}$ .

**$c_k$  update:** In this variant we use the proximity-only update (63). The residual gradient w.r.t.  $c_k$  involves  $\nabla_{c_k} \Phi(x_i - c_k)$ , which is a tensor of polynomial derivatives. Deriving the closed-form is possible but algebraically heavy and we defer it. This makes the local-EDMD M-step a GEM step for  $c_k$ , analogous to Variant A.

## 9 Model Selection

### 9.1 How many clusters?

Given the model with  $N$  clusters, we want to compare models with different  $N$ . The Bayesian approach computes the **marginal likelihood**  $p(X, F \mid N)$ —the probability of the data under model  $N$  after integrating out all parameters.

### 9.2 The NIW marginal likelihood (derivation)

For a single cluster  $k$  with points  $\{x_i : z_i = k\}$ , the marginal likelihood integrates out  $(c_k, \Sigma_k)$ :

$$p(X_k) = \int p(X_k \mid c_k, \Sigma_k) p(c_k, \Sigma_k) dc_k d\Sigma_k. \quad (78)$$

Because the NIW prior is conjugate to the Gaussian likelihood, this integral has a closed form:

$$\begin{aligned} \log p(X_k) &= \frac{d}{2} \log \frac{\kappa_0}{\kappa_n} + \frac{\nu_0}{2} \log |\Psi_0| - \frac{\nu_n}{2} \log |\Psi_n| \\ &\quad + \log \Gamma_d\left(\frac{\nu_n}{2}\right) - \log \Gamma_d\left(\frac{\nu_0}{2}\right) - \frac{R_k d}{2} \log \pi, \end{aligned} \quad (79)$$

where  $\Gamma_d$  is the multivariate gamma function  $\Gamma_d(a) = \pi^{d(d-1)/4} \prod_{j=1}^d \Gamma(a + (1-j)/2)$ .

In our model, we add the residual penalty term:

$$\log p(X_k, F_k) = \log p(X_k) - \frac{1}{2\sigma^2} \sum_{i: z_i=k} \|\varepsilon_k(x_i)\|^2. \quad (80)$$

The total log marginal is:

$$\log p(X, F | N) = \underbrace{\log \frac{\Gamma(N\alpha_0)}{\Gamma(P + N\alpha_0)} \prod_k \frac{\Gamma(R_k + \alpha_0)}{\Gamma(\alpha_0)}}_{\text{Dirichlet-Categorical marginal}} + \sum_k \log p(X_k, F_k). \quad (81)$$

The model with the highest  $\log p(X, F | N)$  is preferred.

## 10 Summary: What Each Piece Does

| Object              | What it controls                            | How it's determined                                                            |
|---------------------|---------------------------------------------|--------------------------------------------------------------------------------|
| $\pi_k$             | Prior probability of cluster $k$            | MAP with Dirichlet prior (Eq. 35)                                              |
| $c_k$               | Center of cluster $k$                       | MAP with NIW prior + residual gradient (Eq. 71)                                |
| $\Sigma_k$          | Shape/size of cluster $k$                   | NIW posterior mode (scatter around $c_k$ )                                     |
| $\sigma^2$          | Balance: proximity vs. residual             | Auto-calibrated from initial residuals; fixed during EM                        |
| $L_k$ (local model) | Vector-field prediction in cluster $k$      | Depends on variant (analytic $\mathbf{J}$ , LS fit, or EDMD)                   |
| $r_{ik}$            | Soft assignment of point $i$ to cluster $k$ | E-step posterior (Eq. 57)                                                      |
| $N$                 | Number of clusters                          | Set high initially; pruned by Dirichlet prior; selected by marginal likelihood |

Table 1: Summary of all quantities in the framework.

| Hyperparameter               | Meaning                     | Our choice                                              |
|------------------------------|-----------------------------|---------------------------------------------------------|
| $\alpha_0$                   | Dirichlet concentration     | 0.5 (sparse-inducing)                                   |
| $\mu_0$                      | Prior mean for centers      | Data mean $\bar{x}$                                     |
| $\Lambda_0$ (or $\kappa_0$ ) | Prior precision for centers | 0.01 $I$ (weak: centers free to move)                   |
| $\Psi_0$                     | Prior scale for covariances | 10 $I$ or 1 $I$ (moderate)                              |
| $\nu_0$                      | Prior df for covariances    | $d + 2$ (minimal informative)                           |
| $\sigma^2$                   | Residual noise scale        | Auto-calibrated from median residual                    |
| $\lambda$ (ridge)            | Regularization for LS/EDMD  | 10 <sup>-6</sup> (Taylor-LS) or 10 <sup>-4</sup> (EDMD) |

Table 2: Hyperparameters and default values.